STATISTICAL LEARNING • ADVANCED SELF-STUDY

CLASSIFICATION: AN APPLIED ROADMAP

ISLR • Chapter 4 | Interpret probabilities, compare classifiers, and validate clinical predictions

How to use this lecture

Part I teaches the mathematics from intuition through derivations, with original graphs, numerical examples and checkpoints. Part II turns the ideas into small, reproducible R commands, with comments explaining what each line is doing. Stop after a checkpoint, solve it, then compare with the answer key.

> **PRACTICE RULE**
>
> Read → work the example by hand → run the R block → verify the output → interpret what changed. The code deliberately avoids loops, custom pipelines, and complex helper functions.

Companion files: a standalone R script and the user-supplied chapter PDF. All teaching text and diagrams in this lecture are original explanations, not reproduced chapter pages.

The reading map

This standalone lecture follows the supplied **ISLR Chapter 4**, printed pp. 129–197: §4.1 overview; §4.2 why not linear regression; §4.3 logistic regression; §4.4 generative models (LDA, QDA, naive Bayes); §4.5 model comparison and KNN; §4.6 generalized linear models including Poisson; §4.7 lab; and the chapter exercises. It extends each explanation with independent worked examples, diagrams, and an `rms`/`Hmisc`-centered R lab.

> **LEARNING OUTCOME**
>
> After this lecture, you should be able to fit and explain an event-risk model, convert log-odds to probability, recognize when discriminant methods may help, quantify the error trade-off, and distinguish a plausible prediction model from a misleadingly high-accuracy classifier.

PART I — THEORY: FUNDAMENTALS THROUGH ADVANCED PRACTICE

1. Classification starts with a different outcome

Chapter anchor: *ISLR §4.1, pp. 129–131.*

Ordinary linear regression predicts a continuous quantity. Classification predicts an outcome category: smoker vs nonsmoker, disease vs no disease, or one of several diagnostic classes. A scientific classifier often has two outputs: a **risk** P(Y=1|X) and a **decision** based on a chosen cutoff. These are different objects.

Suppose Y=1 denotes low birth weight. A model might give estimated risk 0.18. We can report 18% risk without assigning a Yes/No label. At a 0.50 decision threshold, the label is “No”; at a 0.10 threshold, it is “Yes.” The data and predicted risk did not change — only the decision policy did.

$$\hat{Y} = \mathbb{1}\{\hat{p}(X) \geq t\}, \qquad \hat{p}(X) = P(Y = 1 \mid X)$$

> **WORKED EXAMPLE**
>
> **Question.** A patient has a predicted probability 0.32. What are the predictions when the decision cutoff is t=0.50 and t=0.25?
>
> **Solution.** At t=0.50 the class is No (0); at t=0.25 it is Yes (1). Report the probability alongside any class label. The correct cutoff depends on intended decisions, harms, and benefit, not on a universal rule.

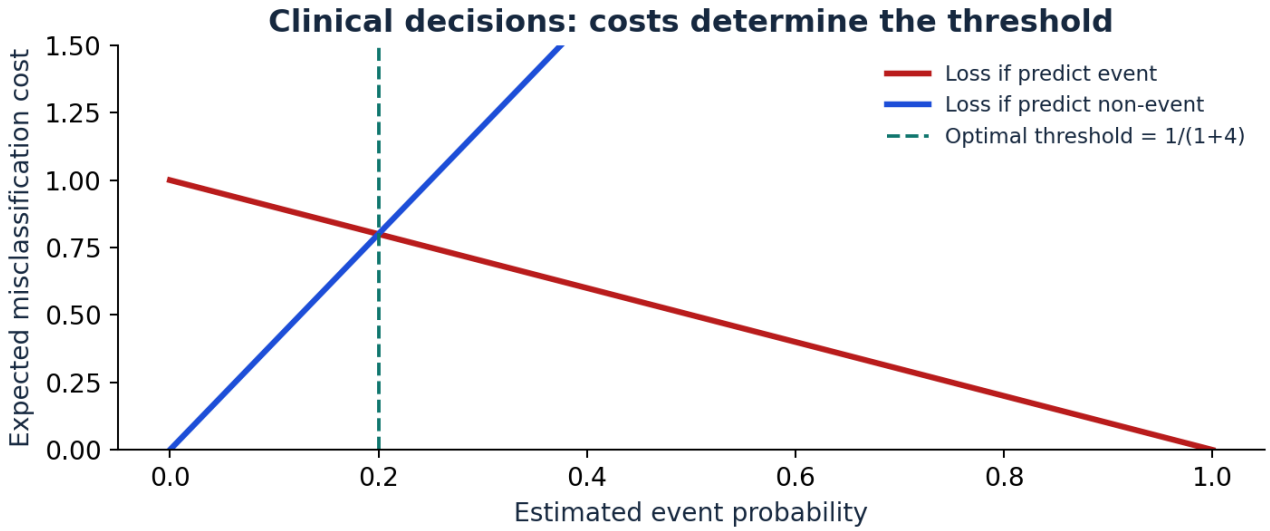


Figure. *Different error costs can justify a probability threshold far below 0.50; this is a decision-analysis issue, not a regression-estimation issue.*

> **MASTERY CHECK 1**
>
> Why is it scientifically clearer to report an estimated probability than only a Yes/No classification? Identify one clinical situation where the threshold might be unusually low.

2. Why not use ordinary least squares?

Chapter anchor: *ISLR §4.2, pp. 131–132; Fig. 4.2.*

A binary outcome can be encoded as 0 or 1, and OLS can regress the indicator on predictors. This is called a linear probability model. It is sometimes a useful approximation, but it has two key weaknesses for generic classification: the fitted value can lie outside [0,1], and the conditional variance of the outcome depends on the event probability, violating constant-variance assumptions of an ordinary homoskedastic linear model.

$$E(Y \mid X) = p(X), \qquad \mathrm{Var}(Y \mid X) = p(X)[1 - p(X)]$$

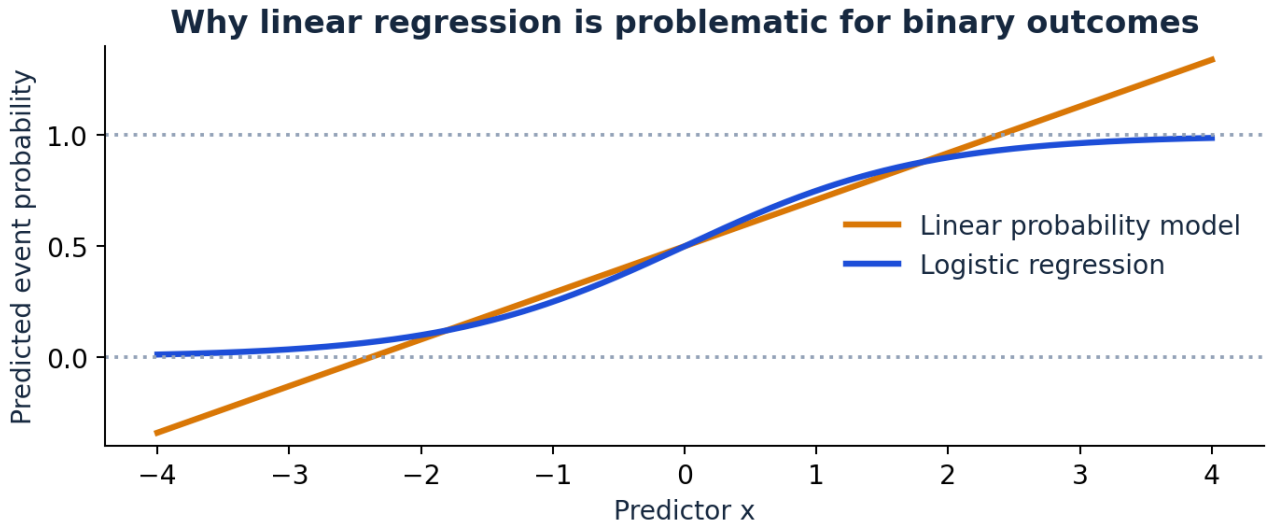


Figure. *For a binary response the linear probability model may leave the valid probability interval; the logistic curve is constrained to the interval.*

For outcomes with more than two unordered classes, numeric encoding imposes an arbitrary order (stroke=1, overdose=2, seizure=3). OLS depends on the numbering and treats label distances as meaningful. A method that works with class probabilities avoids that arbitrary geometry.

WORKED EXAMPLE

Question. A model predicts p=1.15. Can that be interpreted as a 115% chance? What went wrong?

Solution. No. The linear fitted value is not a valid probability. The linear mean structure cannot guarantee a bounded response; a logistic model uses a nonlinear link function to map the real line to (0,1).

MASTERY CHECK 2

Why does Var(Y|X)=p(1−p) change from 0.09 at p=0.1 to 0.25 at p=0.5? How would that affect constant-variance assumptions?

3. Logistic regression: from probability to odds to log-odds

Chapter anchor: *ISLR §4.3.1–4.3.4, pp. 133–139.*

Logistic regression assumes the **log-odds** of the event is a linear function of predictors. An inverse-logit converts that score to probability. The link ensures that fitted risks are between zero and one even for large predictor values.

$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}, \qquad \log \frac{p(x)}{1 - p(x)} = \beta_0 + \beta_1 x$$

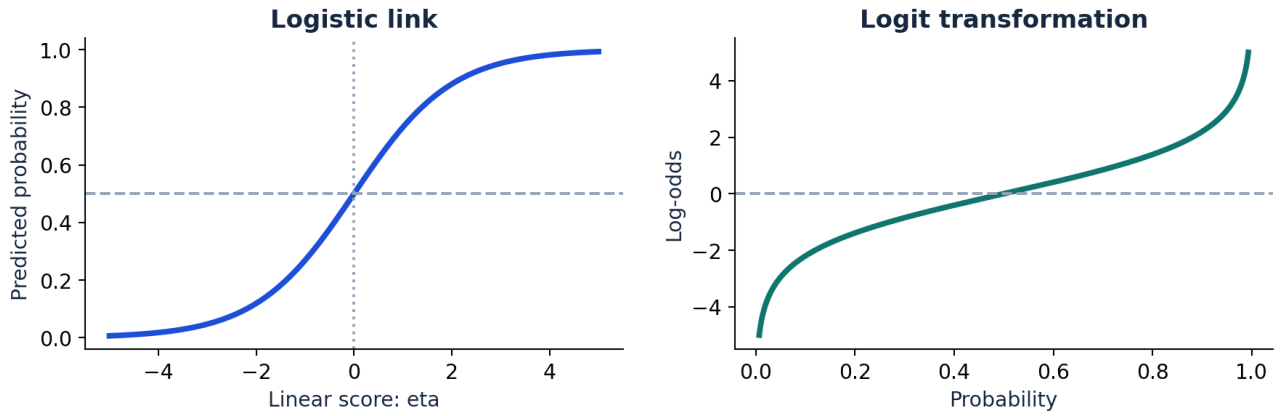


Figure. *Probability and log-odds are not interchangeable. A fixed change in log-odds does not imply a fixed percentage-point change in risk.*

WORKED EXAMPLE

Question. The fitted model is logit(p)=−2+0.5x. Find the event probability when x=2.

Solution. The linear predictor is η=−2+0.5×2=−1. Therefore p=1/(1+e^{1})≈0.269. Increasing x by one multiplies the event odds by exp(0.5)≈1.649, but the risk difference depends on baseline x.

For a multiple logistic model, exp(βj) compares the odds for two observations differing by one unit in Xj *at the same values of all other modeled variables*. A categorical predictor is interpreted relative to its reference level. Odds ratios should not be called risk ratios for common outcomes.

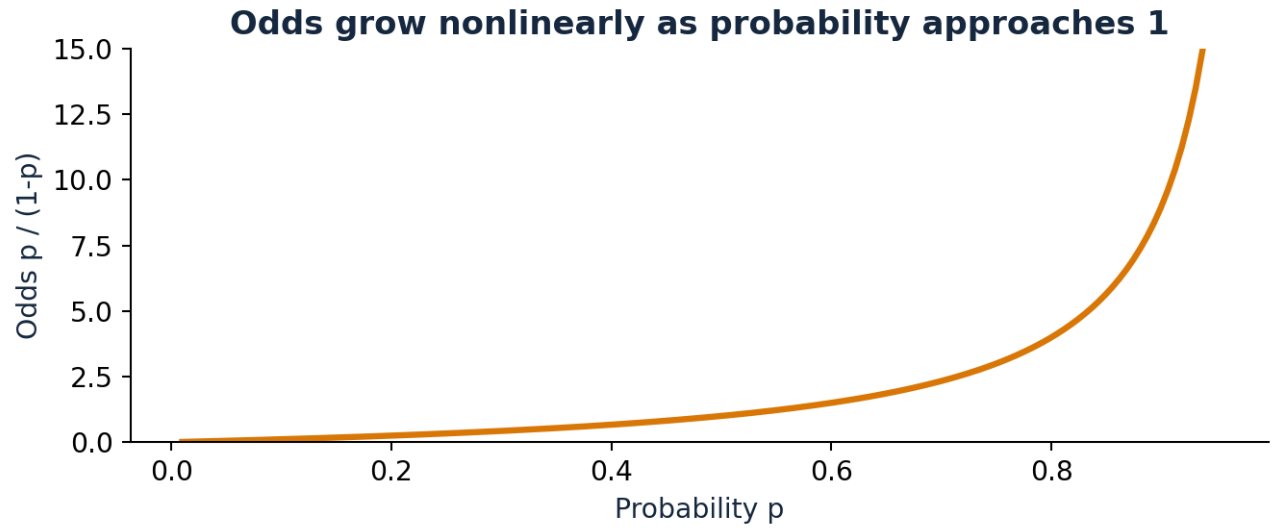


Figure. *An event probability of 0.50 corresponds to odds of 1; a probability of 0.80 corresponds to odds of 4.*

MASTERY CHECK 3

A binary predictor has β=0.7. Compute its odds ratio. Why can you not report the associated absolute change in risk without a baseline probability?

4. Estimation: maximum likelihood, confidence intervals, and model size

Chapter anchor: *ISLR §4.3.2–4.3.3, pp. 135–137.*

Unlike OLS, logistic regression normally estimates its coefficients by maximum likelihood: choose parameters that make the observed sequence of events and nonevents most probable. The likelihood for a binary observation is $p_i^{y_i}(1-p_i)^{(1-y_i)}$. Software maximizes the sum of its logarithm. A Wald statistic divides a coefficient by its estimated standard error, but p-values can be unstable with rare events, separation, small cells, or heavy model selection.

$$L(\beta) = \prod_i p_i^{y_i}(1-p_i)^{1-y_i}$$

Confidence intervals for an odds ratio transform the coefficient interval by exponentiation. If βj=0.4 and SE=0.2, the illustrative Wald interval is 0.4±1.96×0.2=[0.008,0.792], giving an odds-ratio interval about [1.008,2.208]. This approximation should not be treated as universally reliable in small sparse data.

> **WORKED EXAMPLE**
>
> **Question.** A marker has β=−0.35, SE=0.15. Give its adjusted OR and approximate 95% Wald CI.
>
> **Solution.** OR=exp(−0.35)≈0.705. Coefficient CI=−0.35±1.96×0.15=[−0.644,−0.056]. Exponentiating gives OR CI approximately [0.525,0.946]. This is an association conditional on model covariates, not automatically a causal effect.

5. Confounding, adjustment, and the question you actually asked

Chapter anchor: *ISLR §4.3.4 and Fig. 4.3, pp. 137–139.*

A predictor can appear associated with the outcome in a univariable model yet weaken, strengthen, or reverse when related predictors enter. In the chapter's *Default* example, student status and credit-card balance are related; adjustment changes interpretation of student status. An adjusted odds ratio compares individuals at a common set of covariate values; it does not repair unmeasured confounding or incorrect time ordering.

For clinical phenotyping, define the index date, predictors available before prediction, and exactly when the outcome is observed. Using a post-outcome measurement or a feature that directly defines the diagnosis is leakage, not successful prediction.

> **WORKED EXAMPLE**
>
> **Question.** A variable has a large unadjusted OR but a smaller adjusted OR after adding a correlated exposure. Does this prove confounding?
>
> **Solution.** No. The change may reflect confounding, non-collapsibility of odds ratios, different model specifications, selection, or sampling error. Examine a causal diagram, measurement timing, and whether the adjusted contrast answers the intended scientific question.

> **MASTERY CHECK 4**

> In a low-birth-weight dataset, why must measured birth weight in grams be excluded as a predictor of the binary “low birth weight” outcome?

6. More than two classes: multinomial logistic regression

Chapter anchor: *ISLR §4.3.5, pp. 139–141.*

Binary logistic regression generalizes to K unordered outcome classes through a reference class and the softmax. For each of K−1 non-reference classes, we fit a log odds score relative to the reference. Predicted probabilities are nonnegative and add to one. This is different from fitting K unrelated linear regressions to arbitrary numeric labels.

$$P(Y = k \mid x) = \frac{e^{\eta_k(x)}}{1 + \sum_{j=1}^{K-1} e^{\eta_j(x)}}$$

WORKED EXAMPLE

Question. Three diagnostic classes have scores 0, log(2), and log(3). Find their probabilities.

Solution. Exponentiated scores are 1,2,3, total 6; probabilities are 1/6,2/6,3/6. The labels are mutually exclusive and their probabilities sum to one.

7. Generative classification: Bayes and class-specific distributions

Chapter anchor: *ISLR §4.4, pp. 141–142.*

Logistic regression models P(Y|X) directly. Generative methods model a prior πk=P(Y=k) and the conditional distribution fk(x)=P(X=x|Y=k), then apply Bayes’ theorem. That extra distributional structure can help when the classes are approximately well described by the chosen models, but it can hurt when assumptions are badly wrong.

$$P(Y = k \mid x) = \frac{\pi_k f_k(x)}{\sum_j \pi_j f_j(x)}$$

WORKED EXAMPLE

Question. A diagnostic event has prevalence 5%. A biomarker value is 10 times more likely in cases than noncases. What is posterior risk?

Solution. Posterior odds = prior odds × likelihood ratio = (0.05/0.95)×10≈0.5263. Posterior probability=0.5263/(1+0.5263)≈0.345. Even a likelihood ratio of 10 need not imply a >50% risk for a rare disease.

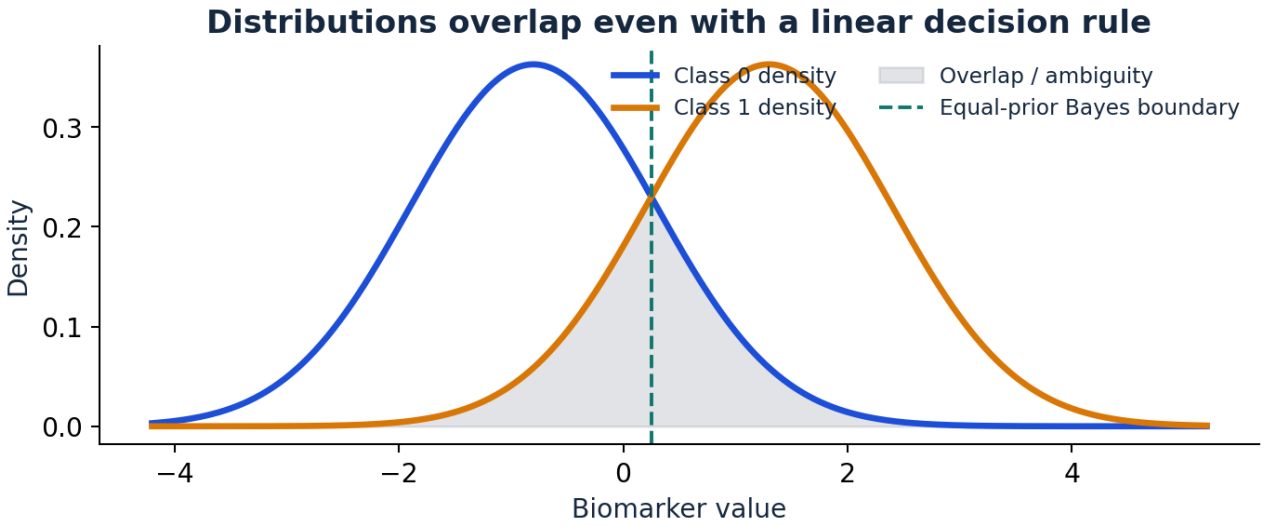


Figure. *With overlapping class-conditional distributions, some errors are unavoidable. Priors shift the decision boundary.*

8. LDA: shared covariance and a linear boundary

Chapter anchor: *ISLR §4.4.1–4.4.2, pp. 142–148.*

Linear discriminant analysis assumes that X|Y=k is approximately multivariate Gaussian with class-specific means but the **same covariance** across classes. When the log posterior densities are compared, shared quadratic terms cancel, leaving a linear classification rule. It estimates priors, class means, and pooled within-class variation.

$$\delta_k(x) = x^\top \Sigma^{-1}\mu_k - \tfrac{1}{2}\mu_k^\top \Sigma^{-1}\mu_k + \log \pi_k$$

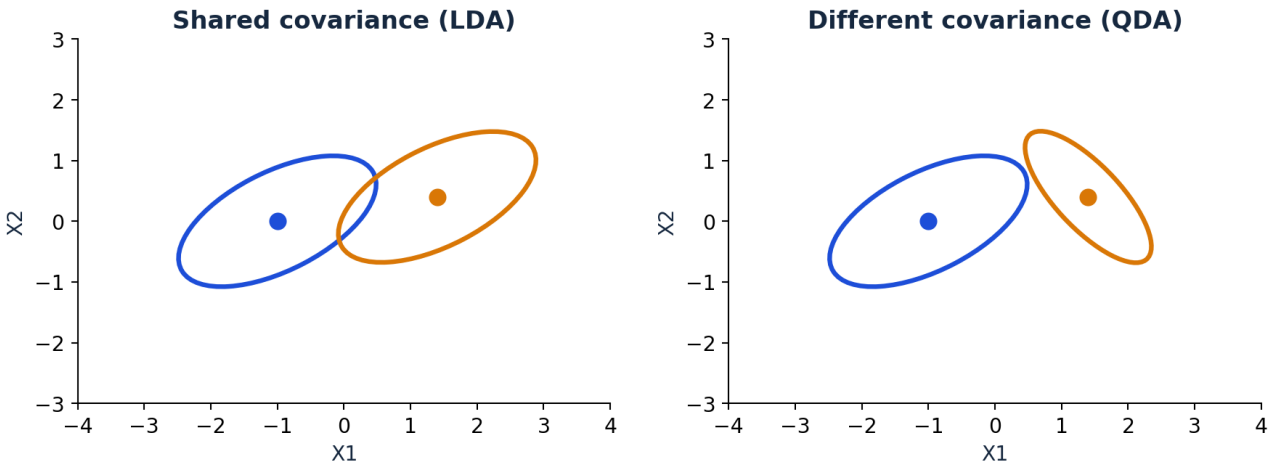


Figure. *Equal covariance describes ellipses with the same spread and orientation, shifted to different centers. Changing the covariance assumptions changes the shape of boundaries.*

LDA can be attractive when data are limited, as its common covariance estimate pools information across classes. It should not be assumed superior to logistic regression merely because both generate a linear boundary. Fit and validation results determine comparative usefulness.

WORKED EXAMPLE

Question. For X|No~N(0,1), X|Yes~N(2,1) and equal priors, classify an observation with x=1.4.

Solution. The equal-prior decision boundary is x=1. Since 1.4>1, classify Yes. A probability model would give a graded posterior rather than only a hard class.

9. Confusion matrices, sensitivity, specificity, and predictive values

Chapter anchor: *ISLR §4.4.2, Tables 4.4–4.7, pp. 148–153.*

A confusion matrix counts **true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN)**. Keep observed status on rows and predictions on columns (or label your axes explicitly). Sensitivity = TP/(TP+FN); specificity = TN/(TN+FP); precision or positive predictive value = TP/(TP+FP). Accuracy can be misleading when events are uncommon.

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \quad \text{Specificity} = \frac{TN}{TN + FP}$$

$$\text{PPV} = \frac{TP}{TP + FP}, \quad \text{Accuracy} = \frac{TP + TN}{N}$$

WORKED EXAMPLE

Question. In 200 patients: TP=32, FN=8, FP=12, TN=148. Compute sensitivity, specificity, PPV, accuracy.

Solution. Sensitivity=32/40=80.0%; specificity=148/160=92.5%; PPV=32/44≈72.7%; accuracy=(32+148)/200=90.0%. An all-negative classifier would score 80% accuracy despite zero sensitivity. Always examine class-specific performance.

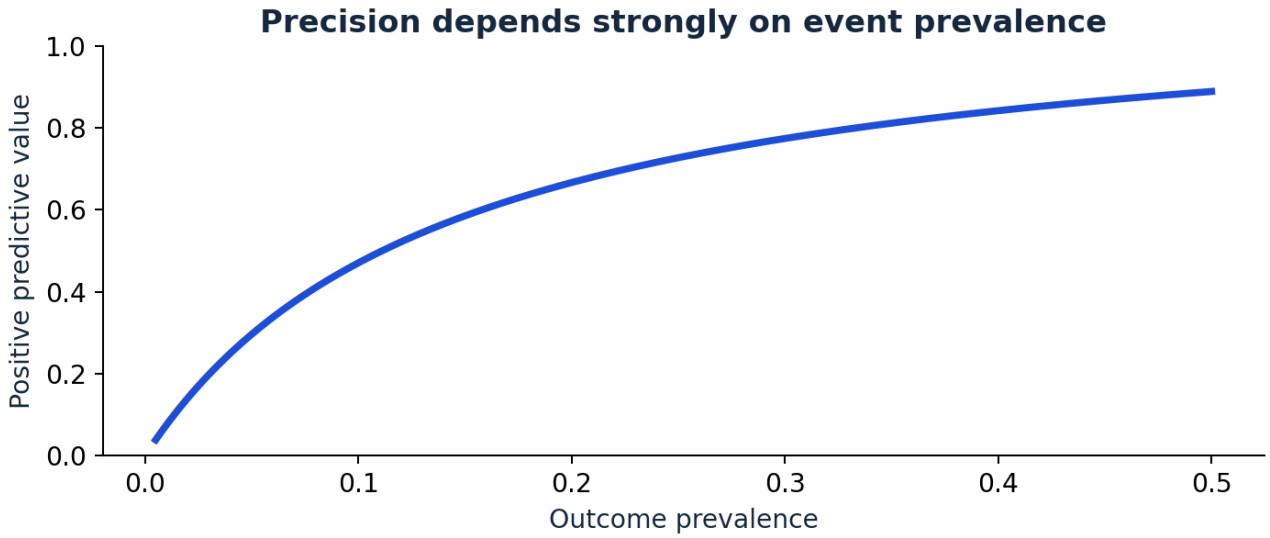


Figure. *Even with unchanged sensitivity and specificity, PPV varies substantially with prevalence. This matters when transporting a model to a new population.*

> **MASTERY CHECK 5**
>
> If an event prevalence drops from 20% to 2% while sensitivity and specificity stay fixed, which predictive value is most immediately at risk of falling?

10. ROC, AUC, calibration, and thresholds

Chapter anchor: *ISLR §4.4.2, Figures 4.7–4.8, pp. 150–151.*

Changing the classification cutoff sweeps out sensitivity (true-positive rate) and false-positive rate (1−specificity). An ROC curve plots these pairs. The area under the ROC curve, or AUC/C-statistic, measures rank discrimination: the probability that a randomly selected case receives a higher score than a randomly selected noncase, accounting for ties in the usual way.

AUC does **not** tell us whether a predicted probability of 0.30 corresponds to 30% observed event risk. That is calibration. The Brier score measures squared probability-prediction error. A method can rank patients well but systematically overestimate everyone's absolute risk.

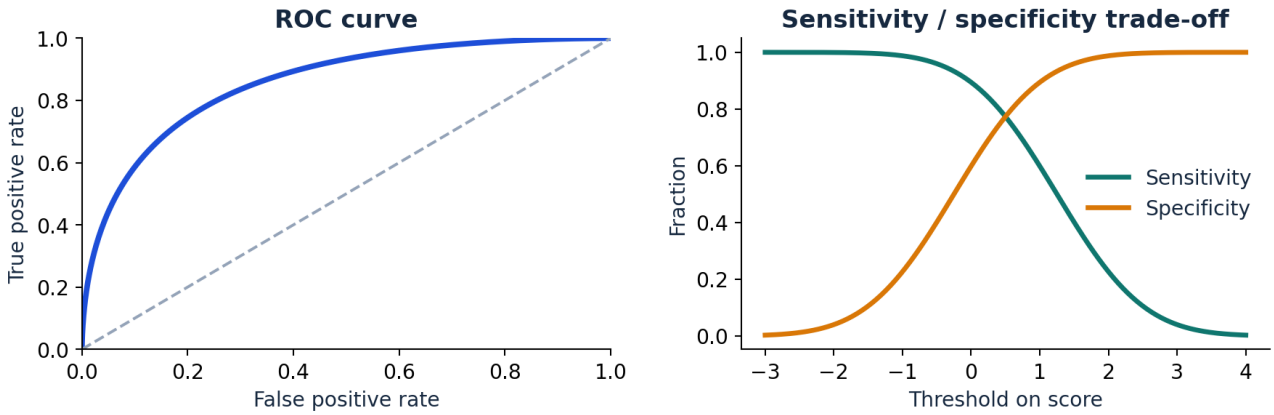


Figure. *AUC evaluates ranking across thresholds, not the clinical optimality of any particular cutoff. The right panel shows the trade-off for one illustrative score model.*

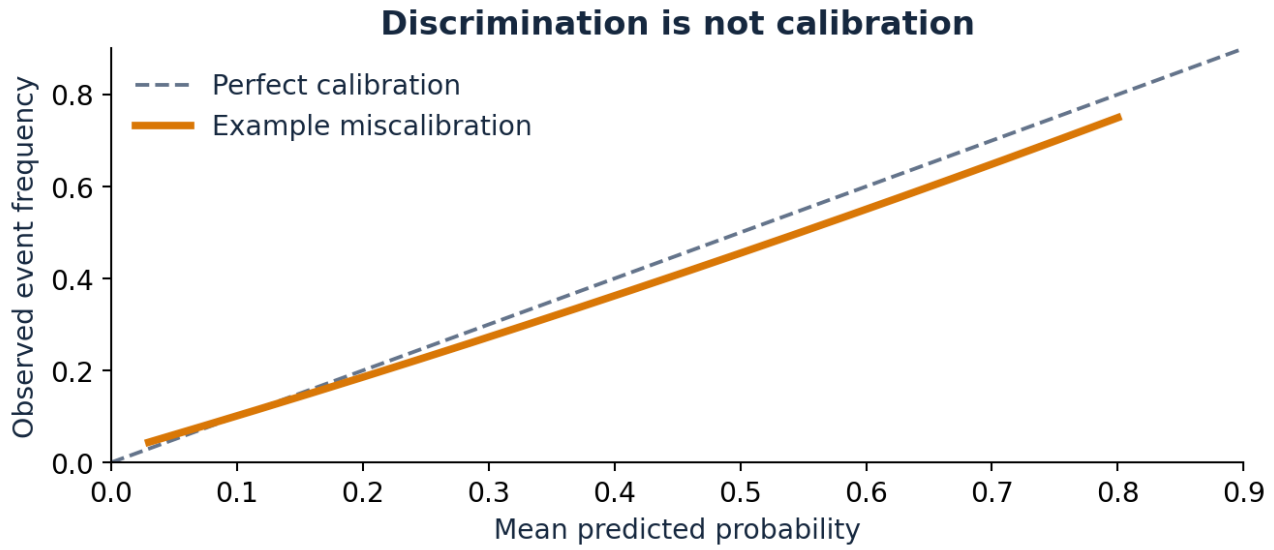


Figure. *Perfect calibration lies along the diagonal. Miscalibration can coexist with strong ROC discrimination.*

WORKED EXAMPLE

Question. A model has AUC 0.84, but patients assigned 30% risk experience the event 12% of the time. What is the central problem?

Solution. It discriminates fairly well but is badly calibrated at that risk level. Consider calibration slope/intercept, optimism, cohort shift, and updating. Do not call it an accurate absolute-risk model solely because AUC is high.

MASTERY CHECK 6

If you change the classification cutoff from 0.50 to 0.20 without changing fitted probabilities, what happens to ROC AUC? What usually happens to sensitivity and specificity?

11. QDA and the flexibility-versus-variance trade-off

Chapter anchor: *ISLR §4.4.3, pp. 152–154.*

Quadratic discriminant analysis (QDA) estimates a different covariance matrix within each class. It can produce nonlinear boundaries even when its Gaussian likelihoods are simple. Compared with LDA, the number of covariance parameters grows rapidly, approximately Kp(p+1)/2; QDA can overfit when p is large relative to the number of cases in each class.

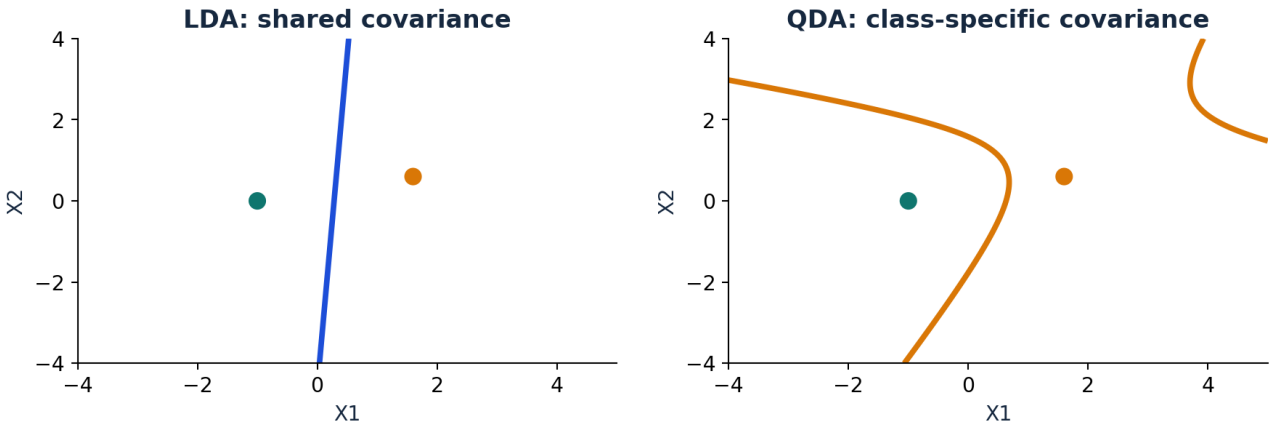


Figure. *With shared covariance, the classifier boundary is linear; allowing separate covariance matrices introduces potential curvature.*

> **WORKED EXAMPLE**
>
> **Question.** You have 10 predictors and 30 cases per class. How many distinct covariance parameters are estimated *per class* by unconstrained QDA?
>
> **Solution.** p(p+1)/2=10×11/2=55 variance/covariance terms per class, plus means and priors. With only 30 observations per class these estimates may be unstable. Simpler LDA or shrinkage might be safer.

12. Naive Bayes: simpler density modeling

Chapter anchor: *ISLR §4.4.4, pp. 154–157.*

Naive Bayes assumes that predictors are conditionally independent *within each class*. That is, fk(x1,...,xp) factors into a product of univariate class-specific density or probability terms. The model can be computationally stable even when high-dimensional multivariate covariance estimation is not feasible. Yet correlated predictors can lead to misleading probability estimates.

$$f_k(x_1, \ldots, x_p) = \prod_{j=1}^{p} f_{kj}(x_j)$$

> **WORKED EXAMPLE**

Question. Two binary test results occur with P(T1=+|case)=0.8 and P(T2=+|case)=0.9. Under the naive conditional-independence assumption, what is P(T1=+,T2=+|case)?

Solution. Multiply 0.8×0.9=0.72. This multiplication requires conditional independence given case status; it is not generally true for correlated measurements from the same biological process.

Naive Bayes is **not** automatically preferable to QDA: the methods make different simplifications. QDA models joint covariance, while naive Bayes discards conditional dependence. Use validation and a scientific understanding of how predictors arise.

MASTERY CHECK 7

Two highly correlated inflammatory markers are both included in naive Bayes. Why might their apparent joint evidence be overstated?

13. A comparative map: logistic, LDA, QDA, naive Bayes, KNN

Chapter anchor: *ISLR §4.5.1–4.5.2, pp. 158–163.*

Logistic regression directly models conditional risk; LDA and QDA construct it through estimated class densities. Naive Bayes uses simpler conditional factorization. K-nearest neighbors (KNN) is local: find the K closest training observations and predict the class majority. KNN makes few parametric shape assumptions, but feature scaling and sample size strongly affect results.

Very small K yields jagged decision boundaries and high variance; large K averages over more observations but can miss local structure. In many predictors, distance becomes less informative unless there is sufficient signal or dimension reduction. The number of neighbors should be chosen by cross-validation *inside the training data*.

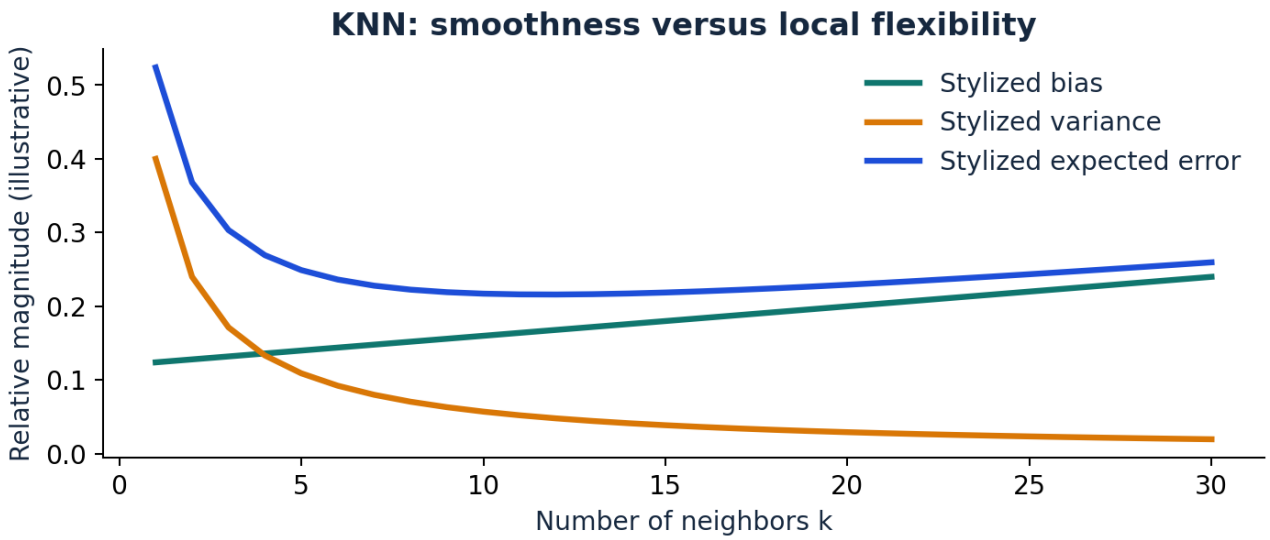


Figure. *Illustrative bias–variance behavior. These are conceptual curves, not measurements on the birth-weight dataset.*

WORKED EXAMPLE

Question. K=5 and the five nearest training subjects include 4 Yes and 1 No. What is the KNN class prediction?

Solution. The majority vote is Yes. A rough vote fraction of 4/5 is not automatically a well-calibrated 80% disease risk; calibration requires separate evaluation.

MASTERY CHECK 8

Why is it harmful to use maternal weight in pounds and age in years as raw Euclidean coordinates without thinking about scale?

14. Generalized linear models: beyond the Bernoulli response

Chapter anchor: *ISLR §4.6, printed pp. 164–171.*

Generalized linear models (GLMs) combine three components: a stochastic response distribution from the exponential family, a linear predictor η=Xβ, and a link function g(μ)=η. OLS assumes normally distributed continuous outcomes with identity link. Logistic regression uses Bernoulli data and the logit link. Poisson regression uses nonnegative integer counts and a log link.

$$g(\mu_i) = x_i^\top \beta, \qquad \operatorname{logit}(p_i) = x_i^\top \beta$$

Poisson regression models counts of events, such as visits in a specified follow-up interval. It assumes E(Y|X)=μ and Var(Y|X)=μ under the basic Poisson model. With a log link, exp(βj) is a multiplicative change in the expected count for a one-unit increase in Xj. If observation time differs, use an **offset log(exposure)** to model rates instead of raw counts.

$$\log E(Y_i \mid x_i) = \beta_0 + \beta^\top x_i + \log(t_i)$$

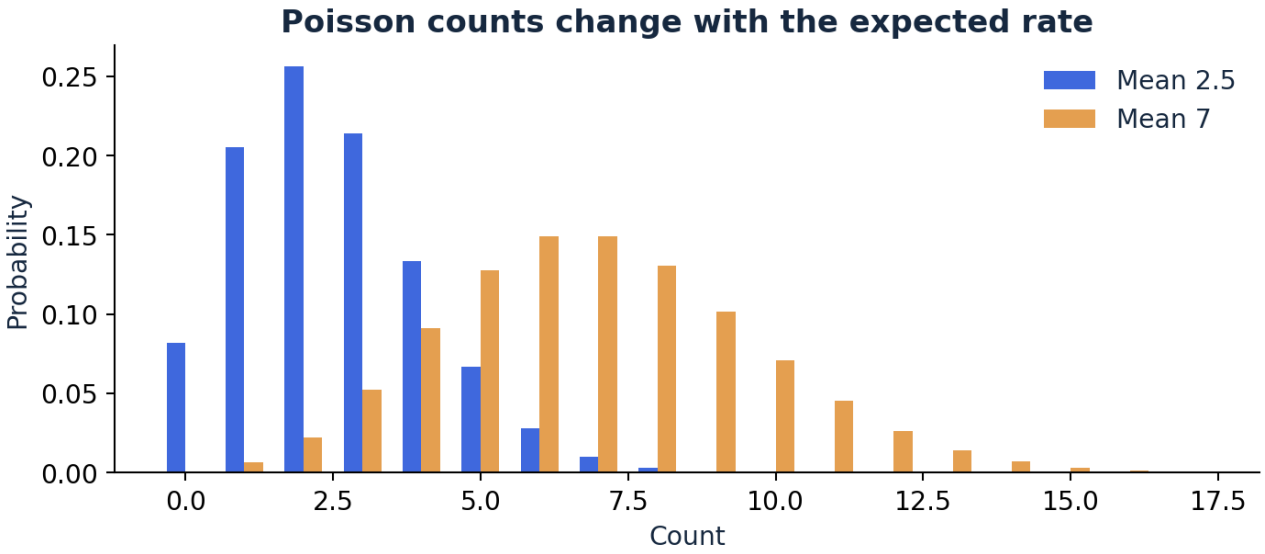


Figure. *Poisson distributions are discrete and become more spread out in absolute count as their means increase.*

WORKED EXAMPLE

Question. Two patients have otherwise identical predictors; one has a log-count coefficient 0.4 for an exposure. What is the expected-count ratio?

Solution. The ratio is exp(0.4)≈1.49. With a time offset and comparable units it can be interpreted as an expected *rate* ratio. Check overdispersion, clustering, and zero inflation instead of applying Poisson uncritically.

15. Advanced research perspective: validation and scientific inference

Training error is not a reliable estimate of error on new patients. A holdout set is useful but noisy in small samples. Cross-validation supports tuning; bootstrap internal validation estimates optimism. External validation across institutions, time, and ancestry is often necessary for scientific transportability. Prediction performance does not imply a predictor has a causal effect, and p-values do not establish a model's clinical usefulness.

THREE QUESTIONS TO KEEP SEPARATE

Association: Does β differ from zero under a specified model? **Prediction:** Does the model rank and calibrate outcomes in new patients? **Decision:** Is acting at a chosen threshold beneficial, given clinical consequences? One statistic cannot answer all three.

PART II — APPLIED R LAB: SIMPLE COMMANDS

This chapter's original lab uses stock-market and credit-default examples. To connect with your earlier linear regression exercises, we instead use `MASS::birthwt`, with `low` as a categorical response. The R script deliberately uses readable individual statements, no for-loops, and no custom programming abstractions. It retains ISLR's methodological sequence.

Lab 1 • Data and leakage checks

Do not include `bwt` in the model predicting `low`; one is mathematically defined from the other.

```
# Packages: install once if missing (not part of repeated analysis)
# install.packages(c("MASS", "Hmisc", "rms", "glmnet", "class"))
library(MASS)
library(Hmisc)
library(rms)

# MASS::birthwt: binary low birth weight is the outcome.
d <- MASS::birthwt
d$low <- factor(d$low, levels = c(0, 1), labels = c("No", "Yes"))
d$race <- factor(d$race, levels = c(1, 2, 3),
                 labels = c("White", "Black", "Other"))
d$smoke <- factor(d$smoke, levels = c(0, 1), labels = c("No", "Yes"))
d$ht <- factor(d$ht, levels = c(0, 1), labels = c("No", "Yes"))
d$ui <- factor(d$ui, levels = c(0, 1), labels = c("No", "Yes"))
d$ptl <- factor(d$ptl > 0, levels = c(FALSE, TRUE),
                labels = c("No", "Yes"))
d$bwt <- NULL       # CRITICAL: bwt defines low, so including it leaks outcome!
d$ftv <- NULL       # Omit unused variable to keep this lab focused.

# Inspect outcome frequency, missingness, and predictors.
Hmisc::describe(d)
table(d$low)
prop.table(table(d$low))
Hmisc::rcorr(as.matrix(d[, c("age", "lwt")]), type = "spearman")
```

```
# Stratified 70/30 split, preserving both outcome levels.
set.seed(2026)
train_no <- sample(which(d$low == "No"),
                   floor(0.70 * sum(d$low == "No")))
train_yes <- sample(which(d$low == "Yes"),
                    floor(0.70 * sum(d$low == "Yes")))
train_id <- sort(c(train_no, train_yes))
train <- d[train_id, ]
test <- d[-train_id, ]
table(train$low)
table(test$low)
```

Lab 2 • Binary logistic regression with glm()

Explain the adjusted smoking OR and note the predictor/reference category. The Wald interval is illustrative; sparse-data inference may require more care.

```
# glm is the base R implementation of a logistic GLM.
fit_glm <- glm(low ~ age + lwt + race + smoke,
               data = train, family = binomial())
summary(fit_glm)

# Odds ratios and Wald confidence intervals are model-based.
exp(coef(fit_glm))
exp(confint.default(fit_glm))

# Probability predictions for participants never used to fit.
p_glm <- predict(fit_glm, newdata = test, type = "response")
head(p_glm)
summary(p_glm)

# Predicted probabilities are not the same as class decisions.
pred_class <- factor(ifelse(p_glm >= 0.50, "Yes", "No"),
                     levels = c("No", "Yes"))
table(Observed = test$low, Predicted = pred_class)
```

Lab 3 • Build a transparent 2 × 2 confusion matrix

Interpret each fraction before looking at its numerical value. Accuracy alone is never sufficient for an imbalanced clinical outcome.

```
# Force the same order so all four cells are traceable.
cm <- table(Observed = test$low, Predicted = pred_class)
cm

TN <- cm["No", "No"]
FP <- cm["No", "Yes"]
FN <- cm["Yes", "No"]
TP <- cm["Yes", "Yes"]

sensitivity <- TP / (TP + FN)
specificity <- TN / (TN + FP)
precision <- TP / (TP + FP)
accuracy <- (TP + TN) / sum(cm)

c(sensitivity = sensitivity, specificity = specificity,
  precision = precision, accuracy = accuracy)
```

Lab 4 • Choose a threshold without refitting

The 0.20 cutoff is solely a teaching example; pre-specify threshold selection and clinical costs in real analysis.

```
# Same probabilities, different decisions.
pred_020 <- factor(ifelse(p_glm >= 0.20, "Yes", "No"),
                   levels = c("No", "Yes"))
pred_050 <- factor(ifelse(p_glm >= 0.50, "Yes", "No"),
                   levels = c("No", "Yes"))

table(Observed = test$low, Predicted = pred_020)
table(Observed = test$low, Predicted = pred_050)

# Discrimination does not change with the cutoff.
Hmisc::somers2(p_glm, as.integer(test$low == "Yes"))
mean((as.integer(test$low == "Yes") - p_glm)^2) # Brier
```

Lab 5 • rms::lrm and Hmisc descriptive analysis

One test concerns the contribution of a modeled variable; the probability curve concerns adjusted prediction. Neither is a causal effect estimate by itself.

```
# rms stores model design information for later inference.
dd <- rms::datadist(train)
options(datadist = "dd")

fit_rms <- rms::lrm(low ~ age + lwt + race + smoke,
                    data = train, x = TRUE, y = TRUE)
print(fit_rms)
rms::anova(fit_rms)
rms::vif(fit_rms)

# Return an adjusted probability curve for maternal weight.
plot(rms::Predict(fit_rms, lwt, fun = plogis),
     xlab = "Maternal weight (lb)",
     ylab = "Estimated probability: low birth weight")

# A likelihood-ratio comparison tests the added term in a GLM.
fit_reduced <- glm(low ~ age + lwt + race,
                   data = train, family = binomial())
anova(fit_reduced, fit_glm, test = "Chisq")
```

Lab 6 • LDA and QDA: class-conditional models

LDA assumes shared covariance, QDA estimates class-specific covariance. The test AUC is unstable with a small holdout.

```
# Start with continuous predictors only for clear covariance models.
lda_fit <- MASS::lda(low ~ age + lwt, data = train)
qda_fit <- MASS::qda(low ~ age + lwt, data = train)

lda_result <- predict(lda_fit, newdata = test)
qda_result <- predict(qda_fit, newdata = test)

lda_fit$prior
lda_fit$means
lda_fit$scaling

head(lda_result$posterior)
table(Observed = test$low, Predicted = lda_result$class)
table(Observed = test$low, Predicted = qda_result$class)

Hmisc::somers2(lda_result$posterior[, "Yes"],
               as.integer(test$low == "Yes"))
Hmisc::somers2(qda_result$posterior[, "Yes"],
               as.integer(test$low == "Yes"))
```

Lab 7 • Naive Bayes by hand, two predictors

This simple implementation mirrors the chapter factorization; it is illustrative and may be unstable in high dimensions because multiplying tiny densities can underflow.

```
# Here Gaussian age and weight are assumed independent within class.
yes <- train$low == "Yes"
no <- train$low == "No"

prior_yes <- mean(yes)
prior_no <- mean(no)

score_yes <- prior_yes *
  dnorm(test$age, mean(train$age[yes]), sd(train$age[yes])) *
  dnorm(test$lwt, mean(train$lwt[yes]), sd(train$lwt[yes]))
score_no <- prior_no *
  dnorm(test$age, mean(train$age[no]), sd(train$age[no])) *
  dnorm(test$lwt, mean(train$lwt[no]), sd(train$lwt[no]))

p_naive <- score_yes / (score_yes + score_no)
summary(p_naive)
Hmisc::somers2(p_naive, as.integer(test$low == "Yes"))
```

Lab 8 • KNN, scaling learned from training data only

This is a comparison, not a valid procedure for choosing k: tune k by cross-validation within training data; leave test outcomes untouched until final evaluation.

```
# KNN uses distances; scale both variables using TRAIN means/SDs.
mean_age <- mean(train$age)
sd_age <- sd(train$age)
mean_lwt <- mean(train$lwt)
sd_lwt <- sd(train$lwt)

knn_train <- data.frame(age = (train$age - mean_age) / sd_age,
                        lwt = (train$lwt - mean_lwt) / sd_lwt)
knn_test <- data.frame(age = (test$age - mean_age) / sd_age,
                       lwt = (test$lwt - mean_lwt) / sd_lwt)

knn_3 <- class::knn(knn_train, knn_test, cl = train$low, k = 3)
knn_11 <- class::knn(knn_train, knn_test, cl = train$low, k = 11)

table(Observed = test$low, Predicted = knn_3)
table(Observed = test$low, Predicted = knn_11)
mean(knn_3 == test$low)
mean(knn_11 == test$low)
```

Lab 9 • Multivariate model diagnostics and calibration

These displays assess different targets; the residual diagnostic panels are not tests proving that logistic probabilities are accurate.

```
# Examine standard GLM diagnostic displays.
par(mfrow = c(2, 2))
plot(fit_glm, which = c(1, 2, 4, 5))
par(mfrow = c(1, 1))

# Bootstrap optimism correction on the training data.
set.seed(2026)
rms::validate(fit_rms, method = "boot", B = 100)

# On the held-out test data, evaluate calibration and Brier.
p_rms <- plogis(predict(fit_rms, newdata = test,
                        type = "lp"))
y_test <- as.integer(test$low == "Yes")
mean((p_rms - y_test)^2)
Hmisc::somers2(p_rms, y_test)
rms::val.prob(p_rms, y_test, pl = TRUE)
```

Lab 10 • Poisson GLM: counts rather than categories

In an actual cohort with differing follow-up durations, incorporate log(person-time) as an offset. Count ratios become rates only when exposure time is appropriately handled.

```
# MASS::quine contains number of days absent from school.
counts <- MASS::quine
Hmisc::describe(counts)
hist(counts$Days, breaks = 20,
     main = "Days absent", xlab = "Count of days")

poisson_fit <- glm(Days ~ Eth + Sex + Age + Lrn,
                   data = counts, family = poisson())
summary(poisson_fit)
exp(coef(poisson_fit)) # multiplicative expected-count ratios

# Screen for overdispersion; values well above 1 merit attention.
pearson_disp <- sum(residuals(poisson_fit,
                              type = "pearson")^2) /
                df.residual(poisson_fit)
pearson_disp

# Negative binomial provides an alternative for overdispersed counts.
nb_fit <- MASS::glm.nb(Days ~ Eth + Sex + Age + Lrn,
                       data = counts)
summary(nb_fit)
AIC(poisson_fit, nb_fit)
```

Applied interpretation worksheet

- Write the exact meaning of Y=Yes. Verify bwt was removed because low is defined using bwt.
- Interpret one odds ratio, compute a 95% interval, then calculate two adjusted probabilities at clinically meaningful covariate settings.
- Report sensitivity, specificity, PPV, event prevalence, and AUC on data not used to fit the model.
- Describe how QDA and KNN increase flexibility relative to LDA, and why extra flexibility may hurt small datasets.
- Distinguish a Poisson count ratio from an incidence-rate ratio that includes an exposure-time offset.

Answer key — theory checks

1. A probability preserves risk gradation and supports different justified thresholds; a hard label discards that information.

2. Bernoulli variance is $0.1\times0.9=0.09$ at $p=0.1$ and $0.5\times0.5=0.25$ at $p=0.5$; the variance is not constant over X.

3. $\exp(0.7)\approx2.014$. Absolute risk change depends on the baseline odds and therefore cannot be read directly from β.

4. bwt deterministically defines low birth weight: inclusion would leak the answer and invalidate intended prediction.

5. The positive predictive value often declines sharply as prevalence decreases, even if sensitivity/specificity are unchanged.

6. ROC AUC is unchanged because risk rankings do not change. Usually lowering a threshold raises sensitivity and lowers specificity.

7. Highly correlated conditional features violate naive Bayes factorization and can double-count related signals.

8. A feature measured numerically in larger units may dominate Euclidean distance unless standardization is considered.

Final research checklist and sources

Primary source: uploaded *ISLR Chapter 4*, *Classification*, printed pp. 129–197, including the supplied lab and exercises. Conceptual figures, calculations, R examples using birthwt, and extra clinical interpretation are independently created teaching extensions. The examples are not representations of an externally validated clinical prediction model.

R package documentation: `?glm`, `?MASS::lda`, `?MASS::qda`, `?class::knn`, `?rms::lrm`, `?Hmisc::describe`, `?Hmisc::somers2`, `?MASS::glm.nb`. The code has not been executed in this PDF-authoring environment; run it line by line and inspect warnings before scientific interpretation.